

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 1)

If n is the number of ways five different employees can sit into four indistinguishable offices where any office may have any number of persons including zero, then n is equal to:

- (1) 47
- (2) 53
- (3) 51
- (4) 43

Q2 - 2024 (01 Feb Shift 2)

The lines $L_1, L_2, \dots, L_{20}$ are distinct. For $n = 1, 2, 3, \dots, 10$ all the lines L_{2n-1} are parallel to each other and all the lines L_{2n} pass through a given point P . The maximum number of points of intersection of pairs of lines from the set $\{L_1, L_2, \dots, L_{20}\}$ is equal to :

Q3 - 2024 (27 Jan Shift 2)

Let $\alpha = \frac{(4!)!}{(4!)^{3!}}$ and $\beta = \frac{(5!)!}{(5!)^{4!}}$. Then :

- (1) $\alpha \in \mathbb{N}$ and $\beta \notin \mathbb{N}$
- (2) $\alpha \notin \mathbb{N}$ and $\beta \in \mathbb{N}$
- (3) $\alpha \in \mathbb{N}$ and $\beta \in \mathbb{N}$
- (4) $\alpha \notin \mathbb{N}$ and $\beta \notin \mathbb{N}$

Q4 - 2024 (29 Jan Shift 1)

All the letters of the word "GTWENTY" are written in all possible ways with or without meaning and these words are written as in a dictionary. The serial number of the word "GTWENTY" is

Q5 - 2024 (29 Jan Shift 2)

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Number of ways of arranging 8 identical books into 4 identical shelves where any number of shelves may remain empty is equal to

- (1) 18
- (2) 16
- (3) 12
- (4) 15

Q6 - 2024 (30 Jan Shift 2)

In an examination of Mathematics paper, there are 20 questions of equal marks and the question paper is divided into three sections: A, B and C. A student is required to attempt total 15 questions taking at least 4 questions from each section. If section A has 8 questions, section B has 6 questions and section C has 6 questions, then the total number of ways a student can select 15 questions is _____.

Q7 - 2024 (31 Jan Shift 1)

The total number of words (with or without meaning) that can be formed out of the letters of the word 'DISTRIBUTION' taken four at a time, is equal to _____.

Q8 - 2024 (31 Jan Shift 2)

The number of ways in which 21 identical apples can be distributed among three children such that each child gets at least 2 apples, is

- (1) 406
- (2) 130
- (3) 142
- (4) 136

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Answer Key

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Solutions

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Q1

Total ways to partition 5 into 4 parts are :

$$5, 0, 0, 0 \Rightarrow 1 \text{ way}$$

$$4, 1, 0, 0 \Rightarrow \frac{5!}{4!} = 5 \text{ ways}$$

$$3, 2, 0, 0 \Rightarrow \frac{5!}{3!2!} = 10 \text{ ways}$$

$$2, 2, 0, 1 \Rightarrow \frac{5!}{2!2!2!} = 15 \text{ ways}$$

$$2, 1, 1, 1 \Rightarrow \frac{5!}{2!(1!)^3 3!} = 10 \text{ ways}$$

$$3, 1, 1, 0 \Rightarrow \frac{5!}{3!2!} = 10 \text{ ways}$$

$$\text{Total} \Rightarrow 1 + 5 + 10 + 15 + 10 + 10 = 51 \text{ ways}$$

Q2

$L_1, L_3, L_5, \dots, L_{19}$ are Parallel

$L_2, L_4, L_6, \dots, L_{20}$ are Concurrent

$$\begin{aligned} \text{Total points of intersection} &= {}^{20}C_2 - {}^{10}C_2 - {}^{10}C_2 + 1 \\ &= 101 \end{aligned}$$

Q3

$$\alpha = \frac{(4!)!}{(4!)^{3!}}, \beta = \frac{(5!)!}{(5!)^{4!}}$$

$$\alpha = \frac{(24)!}{(4!)^6}, \beta = \frac{(120)!}{(5!)^{24}}$$

Let 24 distinct objects are divided into 6 groups of 4 objects in each group.

$$\text{No. of ways of formation of group} = \frac{24!}{(4!)^6 \cdot 6!} \in \mathbb{N}$$

Similarly,

Let 120 distinct objects are divided into 24 groups of 5 objects in each group.

No. of ways of formation of groups

$$= \frac{(120)!}{(5!)^{24} \cdot 24!} \in \mathbb{N}$$

Q4

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Solutions

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Words starting with E = 360

Words starting with GE = 60

Words starting with GN = 60

Words starting with GTE = 24

Words starting with GTN = 24

Words starting with GTT = 24

GTWENTY = 1

Total = 553

Q5

3 Shelf empty: (8, 0, 0, 0) → 1 way

$$\left. \begin{array}{l} (7, 1, 0, 0) \\ (6, 2, 0, 0) \\ (5, 3, 0, 0) \\ (4, 4, 0, 0) \end{array} \right\} \rightarrow 4 \text{ways}$$

$$\left. \begin{array}{l} (6, 1, 1, 0) \quad (3, 3, 2, 0) \\ (4, 2, 1, 0) \quad (4, 2, 2, 0) \end{array} \right\} \rightarrow 5 \text{ways}$$

$$\left. \begin{array}{l} (1, 2, 3, 2) \quad (5, 1, 1, 1) \\ (2, 2, 2, 2) \\ (3, 3, 1, 1) \\ (4, 2, 1, 1) \end{array} \right\} \rightarrow 5 \text{ways}$$

Total = 15 ways

Q6

If 4 questions from each section are selected

Remaining 3 questions can be selected either in (1, 1, 1) or (3, 0, 0) or (2, 1, 0)

$$\begin{aligned} \therefore \text{Total ways} &= {}^8C_5 \cdot {}^6C_5 \cdot {}^6C_5 + {}^8C_6 \cdot {}^6C_5 \cdot {}^6C_4 \times 2 + \\ & {}^8C_5 \cdot {}^6C_6 \cdot {}^6C_4 \times 2 + {}^8C_4 \cdot {}^6C_6 \cdot {}^6C_5 \times 2 + {}^8C_7 \cdot {}^6C_4 \cdot {}^6C_4 \\ &= 56 \cdot 6 \cdot 6 + 28 \cdot 6 \cdot 15 \cdot 2 + 56 \cdot 15 \cdot 2 + 70 \cdot 6 \cdot 2 \\ &+ 8 \cdot 15 \cdot 15 \\ &= 2016 + 5040 + 1680 + 840 + 1800 = 11376 \end{aligned}$$

Q7

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Solutions

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We have III, TT, D, S, R, B, U, O, N

Number of words with selection (a, a, a, b)

$$= {}^8C_1 \times \frac{4!}{3!} = 32$$

Number of words with selection (a, a, b, b)

$$= \frac{4!}{2!2!} = 6$$

Number of words with selection (a, a, b, c)

$$= {}^2C_1 \times {}^8C_2 \times \frac{4!}{2!} = 672$$

Number of words with selection (a, b, c, d)

$$= {}^9C_4 \times 4! = 3024$$

$$\therefore \text{total} = 3024 + 672 + 6 + 32 = 3734$$

Q8

After giving 2 apples to each child 15 apples left now 15 apples can be distributed in ${}^{15+3-1}C_2 = {}^{17}C_2$ ways

$$= \frac{17 \times 16}{2} = 136$$

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